Monte Carlo Search and Games - Project Report

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1 Introduction

For this project we applied different Monte Carlo Search algorithm on graphs. The goal was to refute conjecture in the field of spectral graphs. For one of the conjecture we used the algorithms applied to a more restrained class of graphs to search for a new conjecture. We worked on conjecture from [4] and [2].

The Github repository link to our implementation is here : Github link

2 Algorithm used

For this project we implemented and used the following algorithm:

- Nested Monte Carlo Search (NMCS)
- Nested Rollout Policy Adaptation (NRPA)
- Generalized Rapid Action Value Estimation (GRAVE)
- Upper Confidence bound applied to Trees (UCT)

3 Notation

We will consider undirected graph $G = (V, E)$, where V is the set of vertices and E the set of edges. We will consider graph of size $|V| = n$. A is the adjacency matrix with eigen values $\lambda_1 \geq \lambda_2 \geq ... \geq \lambda_n$. D is the diameter.

4 Conjecture and results

4.1 The upper bound of the i-th eigenvalue

This problem from Powers [5] asks :

Is it true that for a graph of order n and for $i = 3, 4$,

$$\lambda_i \leq \lfloor \frac{n}{i} \rfloor$$

For $i = 1$ and $i = 2$ this bound is proven, for $i \geq 5$ it is refuted. We can note that Linz in [3] refuted the conjecture for $i = 4$ using a closed blowup of a graph, which is a transformation of a graph where with parameter t, an integer, we replace each node by a t-clique, and for each edge in the original graph we add an edge between each node of the two cliques. So at the end we get a pretty dense graph with a high number of nodes. From the paper we also know how to compute the eigenvalue of the blowup graph, the formula is given by : $t\lambda_i + t - 1$ for $i = 1, ..., n$, -1 otherwise.

From this formula we get an interesting score function that we can use to compute eigenvalues of blowup graphs without constructing the graph. For this conjecture, the initial state used is a path, the move considered are adding and removing edges, the score function is given by $\lfloor \frac{n}{i} \rfloor - \lambda_i - 1$. If a graph obtained a negative score, then we know that there is an integer t such that the blowup graph of parameter t refute the conjecture. From these we managed to refute the conjecture for $i = 4$ (see figure 1) but not for $i = 3$ with NMCS, NRPA, UCT and GRAVE.

4.2 Spectral gap of triangle-free graphs

This conjecture from Stanic [6] claims that the spectral gap ($\lambda_1 - \lambda_2$) of a connected graph of order n is minimized for some double kite graph.

A double kite graph is a graph with a path where at each end there is complete graph. With the algorithm we didn't manage to refute the conjecture. Instead of refuting the conjecture we used the algorithms on a more restrained class of graphs, triangle-free graphs to see what kind of graph would minimize the spectral gap.

For this problem, the initial state is the path, we only consider adding edges as removing edge and disconnecting the graph would give us a spectral gap of 0 which the algorithm will try to do, the score function is $\lambda_1 - \lambda_2$.

Using NMCS and NRPA gives similar results, the triangle-free graphs that minimize the spectral graph seem to be a double kite where we remove a minimum number of edge so that it does not contain any triangle (see figure 2). We can also remark that for a small number of vertices, this graph is also bipartite, however for a higher number of vertices this might not be true anymore.

4.3 Conjecture on the sum of the two largest eigenvalues

The conjecture (Conjecture 16 from [2]) states the following : If G is a graph on n vertices then $\lambda_1 + \lambda_2 \leq n$.

For this conjecture, the initial state is a random graph on 18 vertices, the moves are adding and removing edges and the score is $\lambda_1 + \lambda_2 - n$. With this setup, NMCS does not manage to find a counter example but NRPA does in a few seconds (see figure 3).

4.4 Conjecture on largest eigenvalue

The conjecture (Conjecture 11 from [2]) states the following : for graphs on $n \geq 5$ vertices we have

$$\lambda_1(G) + \lambda_1(\overline{G}) \leq \frac{4}{3}n - \frac{5}{3} - \begin{cases} f_1(n) & \text{if } n \bmod 3 = 1, \\ 0 & \text{if } n \bmod 3 = 2, \\ f_2(n) & \text{if } n \bmod 3 = 0, \end{cases}$$

where $\lambda_1(\overline{G})$ is the largest eigenvalue of the complement of G and $f_1(n) = \frac{3n-2-\sqrt{9n^2-12n+12}}{6}$ and $f_2(n) = \frac{3n-1-\sqrt{9n^2-6n+9}}{6}$

We did not manage to refute this conjecture, we used different initial states, random graph, paths, random trees, the moves considered were adding and removing edges, and the score function was :

$$\lambda_1(G) + \lambda_1(\overline{G}) - \frac{4}{3}n + \frac{5}{3} + \begin{cases} f_1(n) & \text{if } n \bmod 3 = 1, \\ 0 & \text{if } n \bmod 3 = 2, \\ f_2(n) & \text{if } n \bmod 3 = 0, \end{cases}$$

4.5 Conjecture on energy of a graph

The energy $\mathcal{E}(G) = \sum_{i=1}^{n} |\lambda_i|$. The conjecture from Akbari-Hosseinzadeh [1] states the following: For every graph with maximum degree Δ and minimum degree δ whose adjacency matrix is non-singular,$\mathcal{E}(G) \geq \Delta + \delta$ and the equality holds if and only if G is a complete graph.

The conjecture has been proven to be true for a large number of graphs, and the algorithm does not manage to find a counter example. However if we relax the hypothesis on the non-singularity of the adjacency matrix, NMCS and NRPA finds a counter example easily (see figure 4).

For this conjecture we used random trees on 11 vertices as initial state, we consider both adding and removing edges and the score function is $\mathcal{E}(G) - \Delta - \delta$.

4.6 Lower bound on Diameter and highest eigenvalues

The conjecture states the following: If G is a connected graph on n vertices then $\lambda_1 + D \geq 2 + \sqrt{n-1}$ with D the diameter of the graph (the length of the shortest path between the most distanced nodes). The conjecture has already been refuted. It is from the the paper [2] and is originally from Graffiti.

For this conjecture, we used an initial graph a random graph of 8 vertices with the number of random edges to add for each new node of 4. This will allow the graph to be well connected and have a low diameter, it is therefore closer to the optimal solution which is stated to be a star graph.

To able to use our algorithms, we needed to only remove edges if it kept the graph connected.

We were able to refute this conjecture using UCT, GRAVE and NRPA. However, NMCS failed to refute it (see figure 5).

4.7 Lower bound on Diameter with sum eigenvalues

The conjecture states the following: If G is a connected graph then $D \leq \sum_{\lambda_i > 0} \lambda_i$ with D the diameter of the graph. The conjecture has already been refuted. The conjecture is the 59th conjecture of the Written on the Wall file.

Similarly to the previous conjecture, we only removed edges if it kept the graph connected.

We use random graph of 8 vertices. If we use a path, the algorithm refutes it too quickly. We were able to refute it using all 4 methods (see figure 6).

References

[1] Saieed Akbari and Mohammad Ali Hosseinzadeh. "A short proof for graph energy is at least twice of minimum degree". In: *MATCH Commun. Math. Comput. Chem* 83 (2020), pp. 631–633.

[2] Mustapha Aouchiche and Pierre Hansen. "A survey of automated conjectures in spectral graph theory". In: *Linear algebra and its applications* 432.9 (2010), pp. 2293–2322.

[3] William Linz. "Improved lower bounds on the extrema of eigenvalues of graphs". In: *Graphs and Combinatorics* 39.4 (2023), p. 82.

[4] Lele Liu and Bo Ning. "Unsolved problems in spectral graph theory". In: *arXiv preprint arXiv:2305.10290* (2023).

[5] David L Powers. "Bounds on graph eigenvalues." In: *LINEAR ALGEBRA APPLIC.* 117 (1989), pp. 1–6.

[6] Zoran Stanić. "Graphs with small spectral gap". In: *The Electronic Journal of Linear Algebra* 26 (2013), pp. 417–432.

5 Appendix

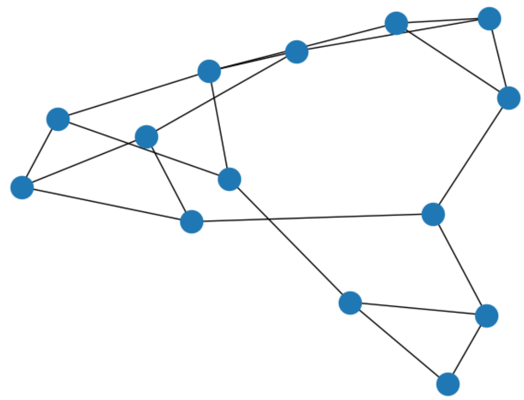

Figure 1: Counter example for conjecture 4.1

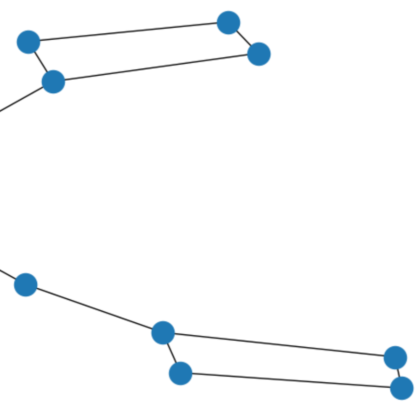

Figure 2: Counter example for conjecture 4.2

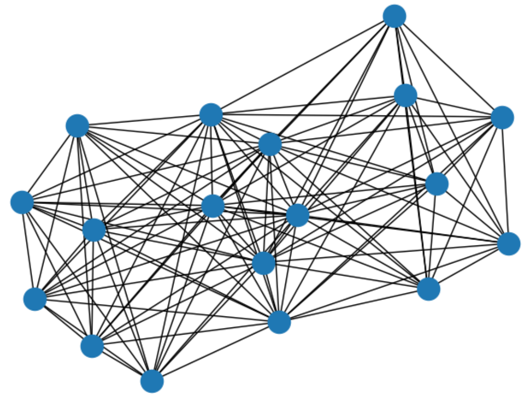

Figure 3: Counter example for conjecture 4.3

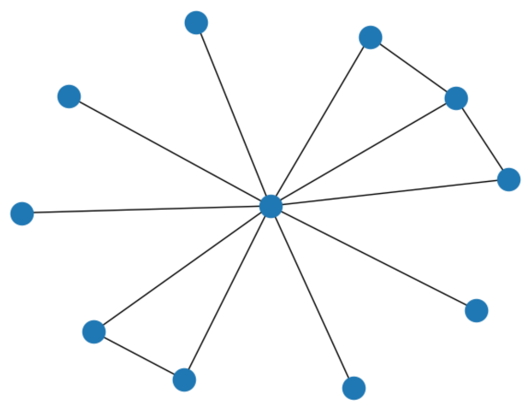

Figure 4: Counter example for conjecture 4.5

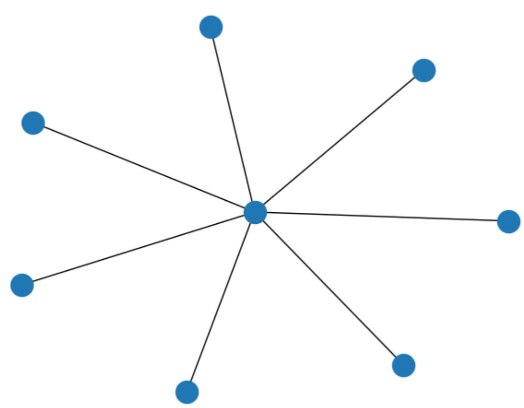

Figure 5: Counter example for conjecture 4.6

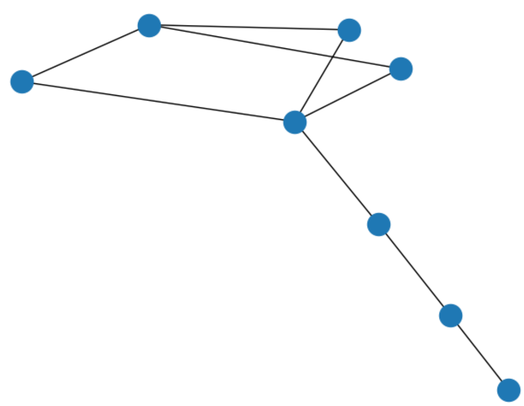

Figure 6: Counter example for conjecture 4.7